

FUNCTIONS OF TESTOSTERONE

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- Responsible for the **distinguishing characteristics of the masculine body**.
- Actions are divided into:
 - a. Functions in fetal life
 - b. Functions in adult life

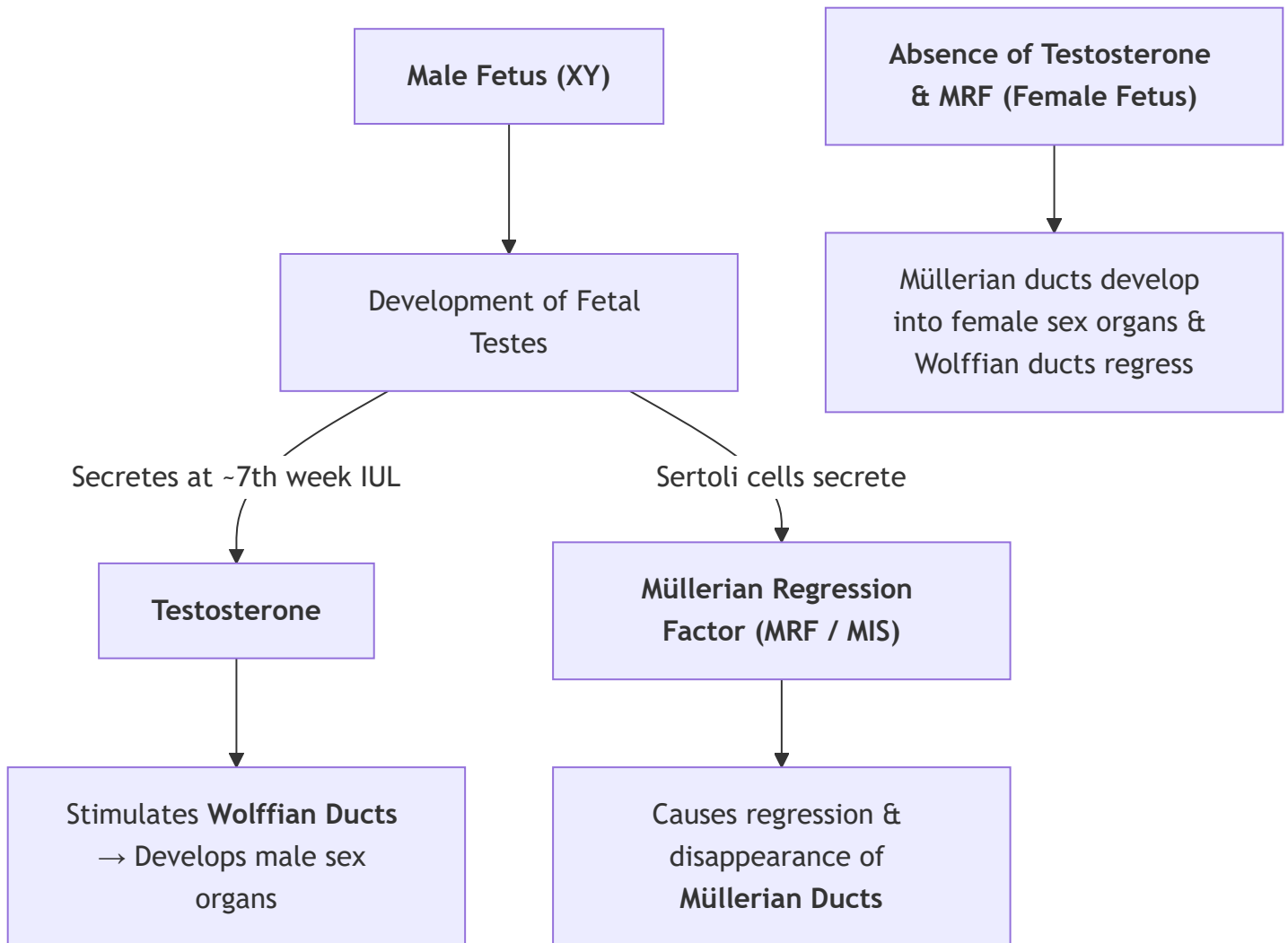
1. FUNCTIONS IN FETAL LIFE

Testosterone performs **3 primary functions** during fetal development:

- a. Sex differentiation in fetus
- b. Development of accessory sex organs
- c. Descent of testes

A. Sex Differentiation in Fetus

- **Determination of Sex** : Determined genetically by sex chromosomes (XY or XX).
- **Differentiation of Sex** : Driven hormonally by **testosterone**.
- The primitive fetus possesses **2 pairs of genital ducts**:
 - a. **Müllerian Ducts (Paramesonephric)** : Give rise to female accessory sex organs (*uterus, fallopian tubes, upper vagina*).
 - b. **Wolffian Ducts (Mesonephric)** : Give rise to male accessory sex organs (*epididymis, vas deferens, seminal vesicles*).



B. Development of Accessory Sex Organs

- Essential for the growth of **external genitalia** (*penis, scrotum*) and **internal genital organs** (*prostate, seminal vesicles, genital ducts*).

C. Descent of Testes

- Process by which the testes migrate from the abdominal cavity into the scrotum, pushed down through the **inguinal canal** just before birth.
- **Hormonal Requirement** : Driven by fetal testosterone.

Cryptorchidism (Undescended Testes) :—

- Congenital disorder characterized by the failure of one or both testes to descend into the scrotum.
- Untreated cases carry a higher risk of developing **testicular cancer** and infertility.
- **Treatment**: Administration of testosterone or gonadotropic hormones (hCG) to stimulate descent.

2. FUNCTIONS IN ADULT LIFE

In adult life, testosterone exerts **2 primary roles**:

- i. **Effect on Sex Organs** : ↑ Increases size of penis, scrotum, and testes after puberty; essential for the **maintenance of spermatogenesis**.
- ii. **Effect on Secondary Sexual Characteristics** : Drives the physical and behavioral features that distinguish males from females after puberty.

SECONDARY SEXUAL CHARACTERS IN MALES

- Physical and physiological changes driven by **testosterone** across target organ systems.

1. Effect on Muscular Growth

- Promotes generalized **development of musculature**.
- **Muscle mass increases by $\approx 50\%$** due to the anabolic effect of testosterone on protein synthesis.

2. Effect on Bone Growth

- After puberty, testosterone ↑ **increases the thickness and density of bones** by:
 - Increasing the deposition of **bone matrix**.
 - Enhancing the deposition of Ca^{2+} (**calcium**) via its protein anabolic activity.
- Causes **early fusion of epiphyses** of long bones with the shaft (diaphysis), eventually limiting further linear growth.

3. Effect on Shoulder & Pelvic Bones

- Promotes **broadening of shoulders**.
- **Specific structural effects on the pelvis :—**
 - a. Lengthening of the pelvis.
 - b. Development of a funnel-like shape.
 - c. Narrowing of the pelvic outlet.

4. Effect on Skin

- ↑ Increases the **thickness of the skin** and ruggedness of subcutaneous tissue.
- ↑ Increases the synthesis of **melanin pigment**, resulting in deepening of skin color.

5. Effect on Hair Distribution

- Stimulates terminal hair growth over the **pubis, along the linea alba up to the umbilicus, on the face (beard/mustache), chest, back, and limbs**.
- In males, pubic hair pattern is distributed with the **base of the triangle directed downwards** and the apex extending upward towards the umbilicus.

6. Effect on Voice

- Causes deepening / **cracking of voice** through:
 - a. Hypertrophy of laryngeal muscles.

- b. Enlargement and lengthening of the larynx.
- c. Thickening of the vocal cords.

7. Effect on Basal Metabolic Rate (BMR)

- ↑ **Increases BMR by about 5 – 10%**, secondary to increased protein synthesis and metabolic activity.

8. Effect on Electrolyte / Water Balance

- ↑ Increases Na^+ **reabsorption** from renal tubules along with secondary water absorption (H_2O retention).
- Leads to a physiological ↑ **increase in Extracellular Fluid (ECF) volume**.

9. Effect on Blood

- Exerts an **erythropoietic action**, causing a mild ↑ **increase in total RBC count** and hemoglobin concentration.